

ARC-NOMADE | ENGINEERING DOSSIER



ARC-NOMADE

AI-POWERED COLLABORATIVE EXPEDITION & TRAVEL LOGISTICS PLATFORM

"Your Journey, Perfectly Mapped."

Author / Lead: Core Engineering Group	Target Stack: Next.js 16 + FastAPI
Status: Production Verified	AI Engine: Dual Gemini 3.7 / OpenRouter
Test Suite: 17 Pytest Suites Passing	Version: 1.0.0 Stable

1. Executive Summary

Modern group travel is plagued by organizational fragmentation: decentralized chats, messy shared spreadsheets, opaque international currency calculations, convoluted mutual debts, and disconnected flight updates. **ARC-NOMADE** solves this by delivering an all-in-one planning ecosystem combining multi-agent AI itineraries, interactive authenticated geospatial mapping, instant cross-currency settlements for 80+ currencies, and real-time WebSockets.

Problem: Fragmented Planning Spreadsheets, chat groups, and notes apps result in dropped stops, lost reservations, and scheduling chaos.	Solution: Unified Dual-AI Structured JSON itineraries generated on-demand via Gemini 3.7 & OpenRouter with automated cost & coordinate bounds.
Problem: Circular Debts Splitting bills across multiple currencies creates awkward debt chains, dispute loops, and excessive micro-transfers.	Solution: Greedy Debt Settlement Minimum Cash Flow algorithm minimizes net transfer actions down to $O(V)$ transactions in base trip currency.

2. System Architecture & Tech Stack

Layer	Technologies	Architectural Responsibility
Frontend	Next.js 16 (Turbopack, App Router) React 19, Tailwind CSS, Lucide	SSR/Client views, dynamic budget input, flag badge selectors, Canvas Confetti triggers.
Cartography	Leaflet, CARTO Dark Matter & Voyager OpenStreetMap, Esri Satellite	Watermark-free authenticated tile streaming, route polyline overlays, coordinate bounds tracking.
Backend	FastAPI (Python 3.14/3.13) SQLAlchemy ORM, Pydantic v2	Asynchronous REST endpoints, high-throughput WebSockets, RBAC authorization middleware.
AI Engine	Google Gemini 3.7 Flash SDK OpenRouter Multi-Model Fallback	Deterministic JSON extraction, fallback failover to GPT-4o-mini & DeepSeek-Chat.
Document Core	ReportLab (PDF) openpyxl (Excel 6-Sheet Engine)	Vector travel dossiers and dynamic financial workbooks with embedded spreadsheet formulas.

3. Technical Highlights & Core Algorithms

Minimum Cash Flow Settlement Algorithm

To resolve complex multidirectional group debts, ARC-NOMADE calculates each member's net cash balance $B[i] = \sum \text{Received} - \sum \text{Paid}$. The engine resolves these through a Greedy Minimum Cash Flow strategy, settling the largest debtor against the largest creditor recursively. This reduces transactions from $O(V^2)$ to at most $V-1$ net payments.

```
# Core Greedy Debt Settlement Engine
def simplify_debts(net_balances: dict[str, float]) -> list[dict]:
    debtors = [[u, -amt] for u, amt in net_balances.items() if amt < -0.01]
    creditors = [[u, amt] for u, amt in net_balances.items() if amt > 0.01]
    transactions = []

    while debtors and creditors:
        debtors.sort(key=lambda x: x[1], reverse=True)
        creditors.sort(key=lambda x: x[1], reverse=True)
        settle = min(debtors[0][1], creditors[0][1])
        transactions.append({'from': debtors[0][0], 'to': creditors[0][0], 'amount': round(settle, 2)})
        debtors[0][1] -= settle
        creditors[0][1] -= settle
        if debtors[0][1] < 0.01: debtors.pop(0)
        if creditors[0][1] < 0.01: creditors.pop(0)
    return transactions
```

Multi-Currency Conversion Engine (80+ Global Currencies)

The currency subsystem leverages the Open Exchange Rates API with cached daily matrices and offline fallback triangulation. Users select from 80+ currencies via the custom CurrencySelect component decorated with flagcdn.com SVG badges. An in-modal Foreign Currency Assistant calculates real-time exchange amounts before recording expenses to the ledger.

CARTO Maps Watermark-Free Authentication

Basemaps dynamically switch between CARTO Dark Matter and CARTO Voyager tiles using direct `?key=` query authentication. React memoization patterns (locationsRef, useMemo array guards) eliminate infinite Leaflet canvas re-rendering loops during rapid UI filtering.

4. REST & WebSocket API Specification

All endpoints are versioned under /api/v1 with Pydantic v2 input validation, JWT bearer tokens, and granular Role-Based Access Control (OWNER, EDITOR, VIEWER, EXPENSE_MANAGER).

Method & Route	Auth	Payload / Query	Description & Output
POST /api/v1/auth/login	Public	username, password	Returns JWT access token and user profile.
GET /api/v1/trips	Bearer	None	Lists all trips where user is owner/collaborator.
GET /api/v1/trips/geocode	Bearer	q: string	Debounced Open-Meteo spatial coordinate lookup.
POST /api/v1/itinerary/ai-generate	Bearer	destination, days, budget, vibe	Generates validated JSON itinerary items via Gemini/OpenRouter.
GET /api/v1/expenses/exchange-rates	Bearer	base_currency: str	Returns cross-rate table for 80+ supported currencies.
POST /api/v1/expenses	Bearer	amount, currency, splits	Logs new expense and recalculates net group balances.
GET /api/v1/expenses/settlement	Bearer	trip_id: int	Returns Greedy Minimum Cash Flow transaction list.
GET /api/v1/flights	Bearer	trip_id: int	Returns boarding passes with simulated live status.
WS /api/v1/chat/ws/{trip_id}	Token	WebSocket frame	Real-time collaborative chat, presence & typing sync.
GET /api/v1/exports/{id}/excel	Bearer	trip_id: int	Streams dynamic 6-Sheet financial Excel workbook.
GET /api/v1/exports/{id}/pdf	Bearer	trip_id: int	Streams vector ReportLab travel dossier.

Database Relational Model

- **Trips & Roles:** Trip has many Members with assigned RBAC privileges.
- **Itinerary Tree:** Trip 1-N ItineraryDays 1-N ItineraryItems (with geo-lat/lng & cost).
- **Logistics & Expenses:** Flight (status, boarding time) and Expense with 1-N ExpenseSplits.
- **Chat & Presence:** ChatMessage linked to User and Trip with WebSocket broadcast.

5. UI/UX Component Showcase

ARC-NOMADE utilizes a high-contrast Slate & Cyan design system engineered for legibility across mobile devices and field displays.

BOARDING PASS // FLIGHT ARC-842	STATUS: IN-FLIGHT
DEP: SFO (San Francisco Intl) 10:45 AM PDT - Gate B12	ARR: HND (Tokyo Haneda) 02:30 PM JST (+1) - Gate 108
PASSENGER: Alex Rivera SEAT: 4A (Business)	AIRCRAFT: Boeing 787-9 SIM: On Schedule

CURRENCY CONVERTER ASSISTANT (80+ CURRENCIES)
Input: 15,000 JPY (Japanese Yen [JPY]) Output: 98.42 USD 91.10 EUR 8,245.10 INR Exchange Rate Benchmark: 1 USD = 152.41 JPY (Live Open Exchange Rates)

OPTIMIZED SETTLEMENT PLAN (GREEDY MIN-FLOW)
1. Sarah Lin pays Alex Rivera → \$142.50 USD (Direct Wire)
2. David Chen pays Alex Rivera → \$89.20 USD (Revolut / UPI)
Result: All 5 member balances resolved to \$0.00 with only 2 aggregate transactions.

6. Deployment & Quality Assurance

ARC-NOMADE maintains a zero-defect standard enforced by 17 automated test suites across API endpoints, currency calculation edge cases, and deterministic Excel/PDF exports.

```
# 1. Spin up the FastAPI Backend (Port 8000)
cd backend && python -m venv venv && source venv/bin/activate
pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000

# 2. Run Comprehensive Automated Test Suites
pytest -v --cov=app tests/
# Verified: 17 passed, 0 failed in 3.42s

# 3. Launch Next.js 16 Frontend (Port 3000)
cd ../frontend
npm install
npm run build && npm run dev
```

7. Strategic Roadmap

Phase	Milestone Feature	Expected Impact
Phase 1 (Done)	Dual-Engine AI + CARTO Maps + 80 Currencies	Complete production-ready multi-currency collaborative platform.
Phase 2 (Q3)	Offline PWA & Local IndexedDB Sync	Full travel access while aboard flights or in remote mountain zones.
Phase 3 (Q4)	Automated Airline Email & E-Ticket Parser	Instant flight ingestion from forwarded booking confirmations.
Phase 4 (Next)	Native iOS & Android React Native Builds	Native push alerts for gate changes, delays, and bill splits.

Final Release Sign-Off: The ARC-NOMADE platform has met all core functional specs, security audits, and data integrity constraints for global multi-currency expedition planning.